Example #1 (Synthetic English):

- $Rule\ r =$ If the child of the first person is the third person, and the parent of the third person is the second person, then the first person is the spouse of the second person.
- $Facts\ F$:
  - f_1**:** The parent of Eve is not Carl.
  - f_2**:** The child of Eve is David.
  - f_3**:** The parent of Carl is Bob.
  - f_4**:** The child of Alice is Carl.
- $Hypothesis\ h_3$: The spouse of Alice is Bob.

The $Context$ is defined as the combined set of facts and rule(s). Both $Context$ and $Hypothesis$ are fed as an input to the model.

Example #1 (Model Input):

- $Context$: The parent of Eve is not Carl. The child of Eve is David. If the child of the first person is the third person, and the parent of the third person is the second person, then the first person is the spouse of the second person. The parent of Carl is Bob. The child of Alice is Carl.
- $Hypothesis$: The spouse of Alice is Bob.

G Rule Overlap Example

After generating the data for every rule in Figure 2, we generate additional examples using combinations of rules. Below, we show how to handle the interaction of two rules: r_2 and r_3. We follow the procedure in Algorithm 1 by generating facts that trigger the rules, but we only take into consideration hypotheses that deal with rule conclusions.

For example, consider the following facts:

Generated Facts

- f_1: `negparent(Eve,Carl)`
- f_2: `child(Eve,David)`
- f_3: `relative(Eve,David)`
- f_4: `predecessor(Eve,David)`

We can generate an example that triggers two rules: f_2 triggers r_2, and f_3 triggers r_3. Feeding the above facts and rules r_2 and r_3 to the reasoner, we obtain the following output (the numbers in parentheses indicate the likelihood of the triple):

$\mathrm{LP}^{\mathrm{MLN}}$ **Reasoner Output O:**

- o_1: `relative(Eve,David) (1.0)`
- o_2: `child(Eve,David) (1.0)`
- o_3: `negparent(Eve,Carl) (1.0)`
- o_4: `spouse(Eve,David) (0.134)`
- o_5: `negspouse(Eve,David) (0.55)`
- o_6: `predecessor(Eve,David) (1.0)`

We produce hypotheses that trigger both rules together. For example, here we generate two hypotheses coming from o_4 and o_5. The confidence (weight) of a hypothesis is given by the $\mathrm{LP}^{\mathrm{MLN}}$ reasoner. Taking o_5 as a hypothesis, we feed the following example to the model:

Example #2 (Model Input):

- $Context$: The parent of Eve is not Carl. The child of Eve is David. If the child of the first person is the second person, then the first person is not the spouse of the second person. The relative of Eve is David. If the relative of the first person is the second person, then the first person is the spouse of the second person. The predecessor of Eve is David.
- $Hypothesis$: The spouse of Eve is not David.
- $Weight$: 0.55

We also generate an example, where three rules are triggered: In addition to r_2 and r_3, r_5 is triggered by f_4. We then repeat the same procedure to generate the following example:

Example #3 (Model Input):

- $Context$: The parent of Eve is not Carl. The child of Eve is David. If the child of the first person is the second person, then the first person is not the spouse of the second person. The relative of Eve is David. If the relative of the first person is the second person, then the first person is the spouse of the second person. The predecessor of Eve is David. If the predecessor of the first person is the second person, then the first person is not the spouse of the second person.
- $Hypothesis$: The spouse of Eve is not David.
- $Weight$: 0.6

This procedure is repeated for all combinations of two or more rules. In case when all rules have the same head polarity, we generate a false example by altering the hypothesis and finding the complement of the initial (1-$Weight$) weight. For example, r_2 and r_5 can occur together and both have the same rule head, and thus no conflict occurs. The generated valid example would be as follows:

Example #4 (Model Input):

- $Context$: The parent of Eve is not Carl. The child of Eve is David. If the child of the first person is the second person, then the first person is not the spouse of the second person. The relative of Eve is David. The predecessor of Eve is David. If the predecessor of the first person is the second person, then the first person is not the spouse of the second person.
- $Hypothesis$: The spouse of Eve is not David.
- $Weight$: 0.64

An invalid example is generated from the valid example by altering the hypothesis. Here is an invalid example:

Example #4 (Model Input):

- $Context$: The parent of Eve is not Carl. The child of Eve is David. If the child of the first person is the second person, then the first person is not the spouse of the second person. The relative of Eve is David. The predecessor of Eve is David. If the predecessor of the first person is the second person, then the first person is not the spouse of the second person.
- $Hypothesis$: The spouse of Eve is ~~not~~ David.
- $Weight$: 0.36 (1-0.64)

H Rule Chaining Example

Here is an example that illustrates rule chaining:

Example #5 (Symbolic):

- $Rules\ R =$
  - r_1: `child(A,C)` $\wedge$ `parent(C,B)` $\rightarrow$ `spouse(A,B)`
  - r_2: `child(B,A)` $\rightarrow$ `parent(A,B)`
- $Facts\ F$:
  - f_1: `negparent(Eve,Carl)`
  - f_2: `child(Bob,Carl)`
  - f_3: `child(Alice,Carl)`
- $Hypothesis\ h$: `spouse(Alice,Bob)`

f_2 triggers r_2 which produces $t =$ `child(Bob,Carl)`.

t and f_3 trigger r_1 to validate the hypothesis h. r_1 and r_2 have been chained to validate the hypothesis. Since we used two rules to validate the hypothesis, we say that this is a chain of depth $= 2$.